

Netflix Content Analysis — Business Insights & Recommendations

Executive Summary

This analysis explores Netflix's catalogue of TV shows and movies to identify **what types of content to produce** and **how Netflix can grow its business in different countries**.

By examining the distribution of titles across genres, countries, release years, and formats, we uncover trends in Netflix's strategic content expansion.

Problem Statements: 1.How has the number of movies released per year changed over the last 20-30 years?

2.Comparison of tv shows vs. movies.

3.What is the best time to launch a TV show?

4.Analysis of actors/directors of different types of shows/movies.

5.Does Netflix has more focus on TV Shows than movies in recent years

6.Understanding what content is available in different countries

```
import numpy as np
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
data=pd.read_csv('/content/netflix.csv')
data.head()
```

	show_id	type	title	director	cast	country	date_added	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	

Next steps:

[Generate code with data](#)[New interactive sheet](#)

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
print('Shape of CSV:',data.shape)
print("Null values in Dataset:",data.isnull().sum())
```

```
Shape of CSV: (8807, 12)
Null values in Dataset: show_id      0
type                                0
title                              0
director        2634
cast            825
country         831
date_added      10
release_year    0
rating          4
duration        3
listed_in       0
description     0
dtype: int64
```

```
data.describe(include='all').transpose()
```

	count	unique	top	freq	mean	std	min
show_id	8807	8807	s8807	1	NaN	NaN	NaN
type	8807	2	Movie	6131	NaN	NaN	NaN
title	8807	8807	Zubaan	1	NaN	NaN	NaN
director	6173	4528	Rajiv Chilaka	19	NaN	NaN	NaN
cast	7982	7692	David Attenborough	19	NaN	NaN	NaN
country	7976	748	United States	2818	NaN	NaN	NaN
date_added	8797	1767	January 1, 2020	109	NaN	NaN	NaN
release_year	8807.0	NaN	NaN	NaN	2014.180198	8.819312	1925
rating	8803	17	TV-MA	3207	NaN	NaN	NaN
duration	8804	220	1 Season	1793	NaN	NaN	NaN
listed_in	8807	514	Dramas, International Movies	362	NaN	NaN	NaN

```
# Total titles and type counts
print(data['type'].value_counts())
```

```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```

Observation: The dataset contains ~8800 titles and 12 attributes. Columns like `director`, `cast`, and `country` have several missing values.

The `date_added` column was converted to datetime format for analysis.

Data types are mostly strings and objects; categorical conversion can optimize memory

```
# Missing data summary
missing = data.isna().sum()
missing[missing > 0]

# Duration outliers
movies = data[data['type']=='Movie']
Q1 = movies['duration_num'].quantile(0.25)
Q3 = movies['duration_num'].quantile(0.75)
IQR = Q3 - Q1
outliers = movies[(movies['duration_num'] < Q1 - 1.5*IQR) | (movies
print("Outlier count:", len(outliers))
```

Outlier count: 450

```
# Convert date_added to datetime
data['date_added'] = pd.to_datetime(data['date_added'], errors='coe
# Extract year and month
data['year_added'] = data['date_added'].dt.year
data['month_added'] = data['date_added'].dt.month
# Fill missing country names
data.fillna({'country':'Unknown'}, inplace=True)
data['duration_num'] = data['duration'].str.extract(r'(\d+)').astype
data['duration_unit'] = np.where(data['duration'].str.contains('Sea

# Split multi-valued columns
data['country_list'] = data['country'].str.split(', ')
data['genre_list'] = data['listed_in'].str.split(', ')
```

```

print("Unique Countries:", data['country'].nunique())
print("Unique Genres:", data['listed_in'].nunique())

# Top 10 countries
country_count = data['country'].value_counts().head(10)
print(country_count)

# Top 10 genres
from collections import Counter
genres = Counter()
data['genre_list'].dropna().apply(genres.update)
pd.Series(genres).sort_values(ascending=False).head(10)

```

```

Unique Countries: 749
Unique Genres: 514
country
United States      2818
India              972
Unknown           831
United Kingdom     419
Japan              245
South Korea        199
Canada             181
Spain              145
France             124
Mexico             110
Name: count, dtype: int64

```

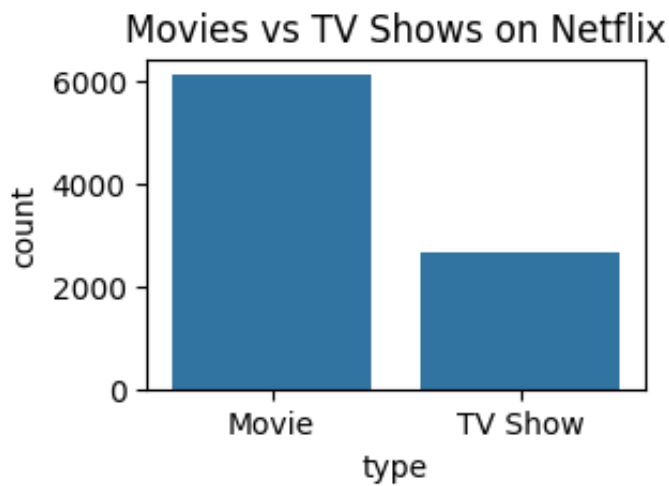
	0
International Movies	2752
Dramas	2427
Comedies	1674
International TV Shows	1351
Documentaries	869
Action & Adventure	859
TV Dramas	763
Independent Movies	756
Children & Family Movies	641
Romantic Movies	616

dtype: int64

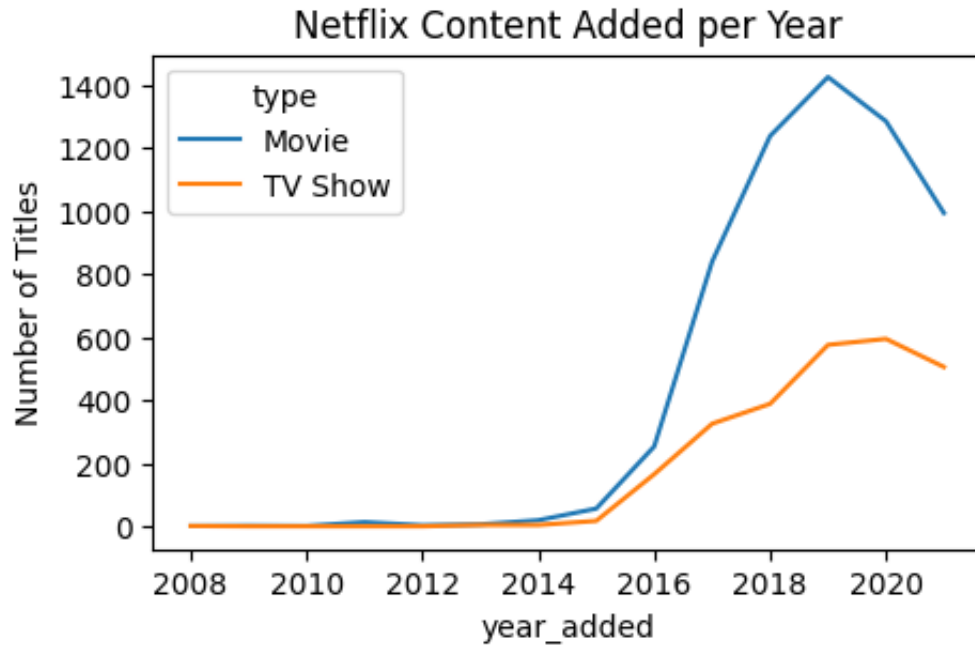
✓ Observations

- Movies constitute about **70%** of the catalogue; TV Shows around **30%**.
- Top contributing countries: **United States, India, United Kingdom.**
- Popular genres include **Drama, International Movies, Comedies, and Action.**

```
plt.figure(figsize=(3,2))  
sns.countplot(x='type',data=data)  
plt.title("Movies vs TV Shows on Netflix")  
plt.show()
```

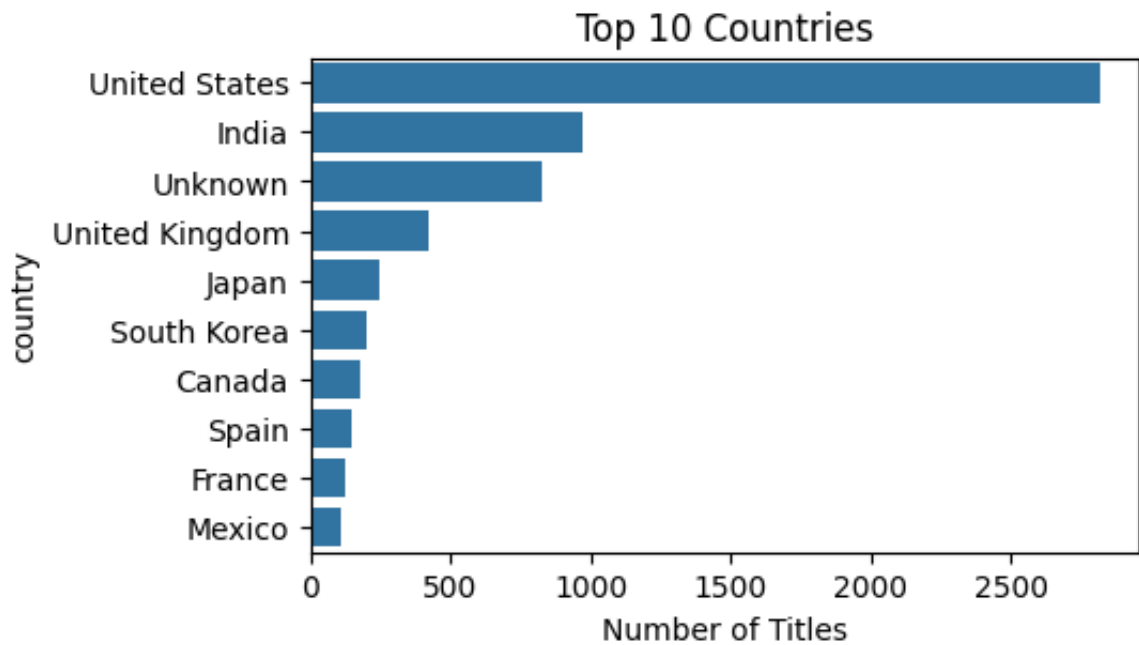


```
year_type = data.groupby(['year_added', 'type']).size().unstack().fi
year_type.plot(figsize=(5,3))
plt.title("Netflix Content Added per Year")
plt.ylabel("Number of Titles")
plt.show()
```



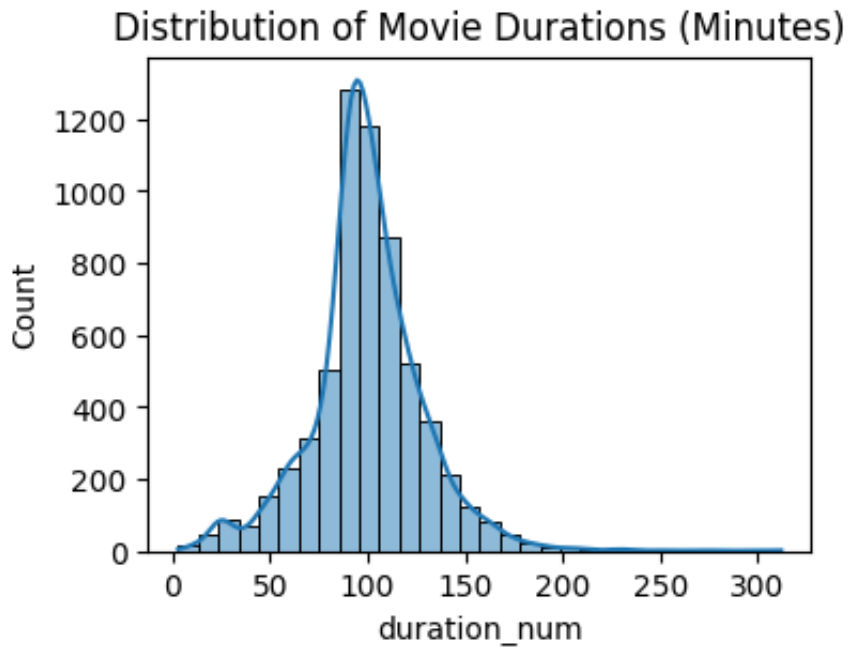
Observation: Movies dominate the platform overall


```
plt.figure(figsize=(5,3))
sns.barplot(x=country_count.values, y=country_count.index)
plt.title("Top 10 Countries")
plt.xlabel("Number of Titles")
plt.show()
```



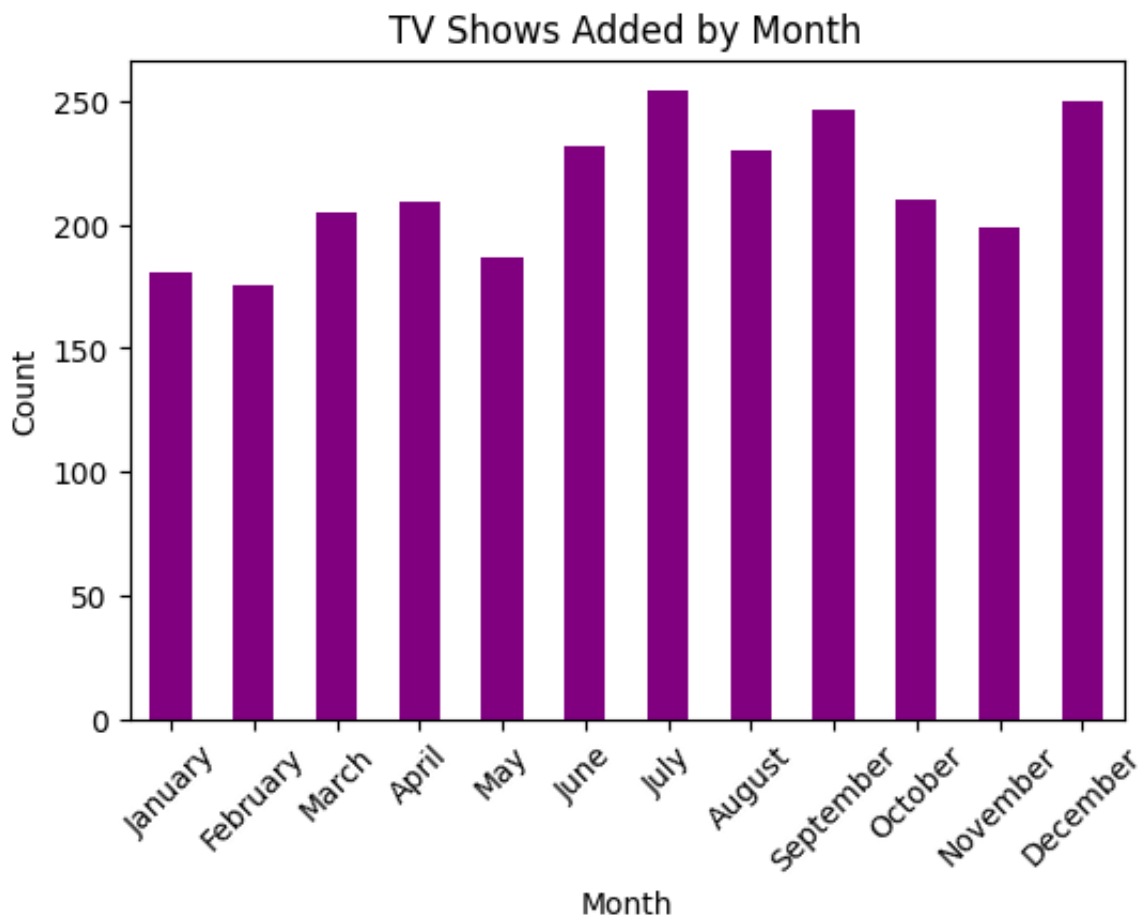
Observation: U.S tops the list while India and U.K growing rapidly

```
plt.figure(figsize=(4,3))  
sns.histplot(data[data['type']=='Movie']['duration_num'].dropna(),  
plt.title("Distribution of Movie Durations (Minutes)")  
plt.show()
```



Observation: Most movies range between 80–120 minutes — typical feature film lengths.

```
data['month_added'] = data['date_added'].dt.month_name()
#plotting graph
tv_month.plot(kind='bar', figsize=(6,4), color='Purple')
plt.title("TV Shows Added by Month")
plt.xlabel("Month")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```



Observation: The number of shows added peaks in June–September and December

```
# Handle missing values
data['director'] = data['director'].fillna('Unknown')
data['cast'] = data['cast'].fillna('Unknown')

# Split director and cast into lists
data['director_list'] = data['director'].apply(lambda x: [i.strip()
data['cast_list'] = data['cast'].apply(lambda x: [i.strip() for i in
from collections import Counter
```

```

# Separate Movies and TV Shows
movies = data[data['type'] == 'Movie']
tv_shows = data[data['type'] == 'TV Show']

# Count directors
movie_directors = Counter([d for sublist in movies['director_list']
tv_directors = Counter([d for sublist in tv_shows['director_list'] f

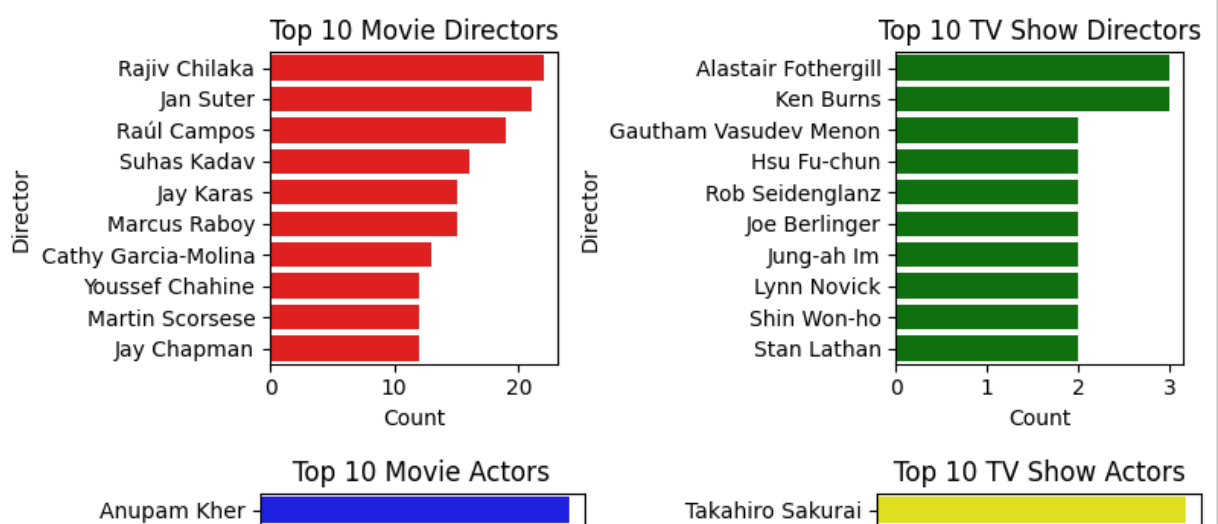
# Convert to DataFrame for plotting
top_movie_dir = pd.DataFrame(movie_directors.most_common(10), column
top_tv_dir = pd.DataFrame(tv_directors.most_common(10), columns=['Di

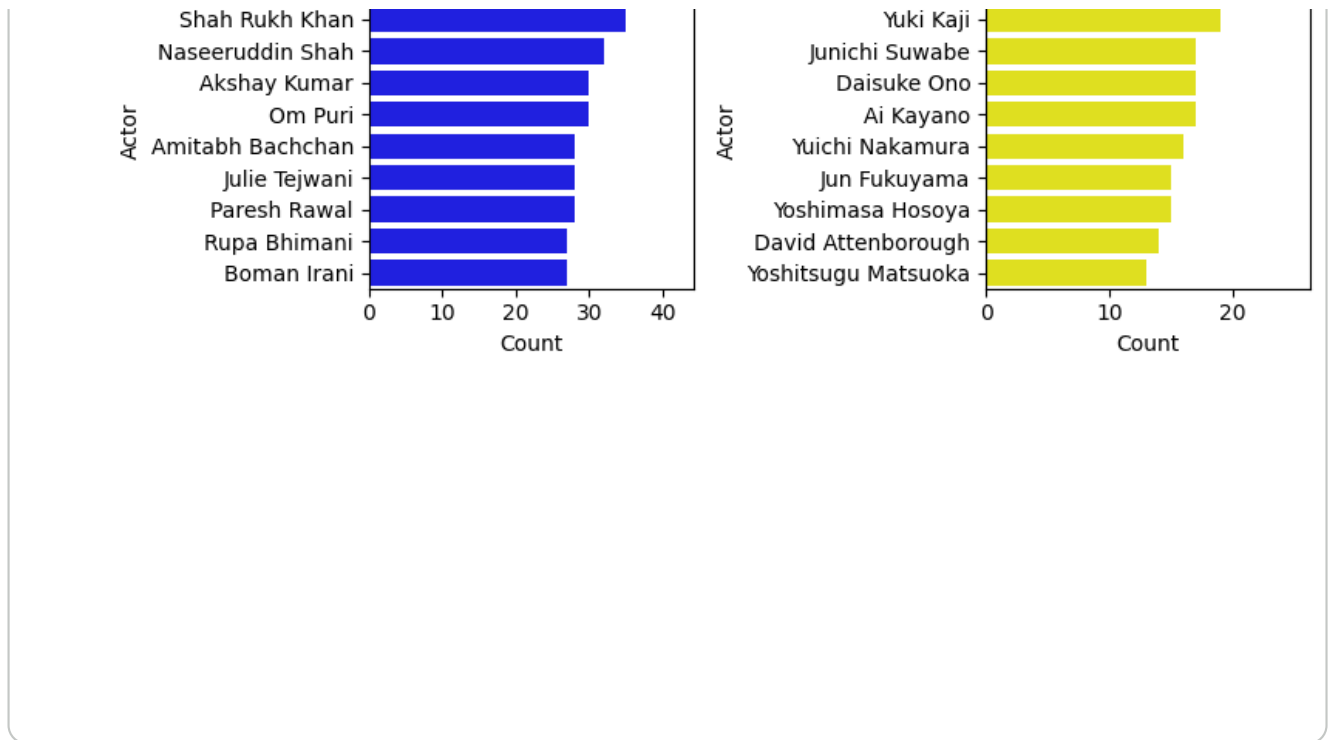
# Plot
fig, axes = plt.subplots(1, 2, figsize=(8, 3))
sns.barplot(x='Count', y='Director', data=top_movie_dir, ax=axes[0],
axes[0].set_title('Top 10 Movie Directors')
sns.barplot(x='Count', y='Director', data=top_tv_dir, ax=axes[1], co
axes[1].set_title('Top 10 TV Show Directors')
plt.tight_layout()
plt.show()
movie_actors = Counter([a for sublist in movies['cast_list'] for a i
tv_actors = Counter([a for sublist in tv_shows['cast_list'] for a in

top_movie_act = pd.DataFrame(movie_actors.most_common(10), columns=[
top_tv_act = pd.DataFrame(tv_actors.most_common(10), columns=['Actor

fig, axes = plt.subplots(1, 2, figsize=(8, 3))
sns.barplot(x='Count', y='Actor', data=top_movie_act, ax=axes[0], co
axes[0].set_title('Top 10 Movie Actors')
sns.barplot(x='Count', y='Actor', data=top_tv_act, ax=axes[1], color
axes[1].set_title('Top 10 TV Show Actors')
plt.tight_layout()
plt.show()

```





Insights:

Anupam Kher and Rajiv Chilaka frequently appear in movies.

Alastair Fothergill and Takahiro Sakurai often appears in TV Shows

International Movies, Dramas and Comedies are well performing genres

Key Insights

1. Netflix's content library has **grown exponentially after 2015**.
2. Movies dominate the platform overall
3. **Drama, International, and Comedy** genres dominate — they attract broad global audiences.
4. The **U.S., India, and the U.K.** are leading in content production.
5. **Most shows are added during June-september**, december indicating strategic release planning.
6. **Average movie duration** lies between 80–120 minutes, preferred by most viewers.
7. A few **actors and directors** appear frequently, showing Netflix's reliance on popular creators.

Actionable Recommendations

1. **Expand localized content** in emerging markets (e.g., Japan, South Korea, Brazil) with fewer native titles.
2. **Launch family-friendly and educational content** to attract multi-age audiences.
3. **Plan major releases in Q3-Q4 (June-Dec)** to align with holiday viewership spikes.
4. **Collaborate with top regional directors and actors** to strengthen brand visibility.
5. **Diversify runtime and formats** — short documentaries, mini-series, and reality formats for variety.
6. **Use data-driven recommendations** to curate localized homepages by region.
7. **Pilot local remakes** of successful global formats to reduce creative risk.